



U.S. General Services
Administration

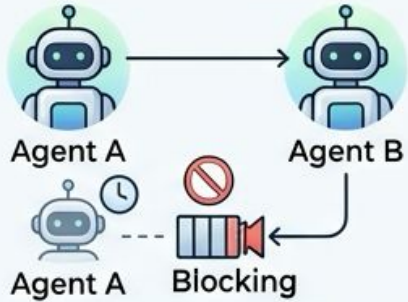


Chapter 4.1: Introduction to AI Agent Deployment and Scaling

US General Service Administration
AI Community of Practice - March 2026

Message Queues

SYNCHRONOUS CALLS (BOTTLENECK)



MESSAGE QUEUES (DECOUPLED)



BROKER-CENTRIC



 RabbitMQ

-  Reliable Delivery
-  Complex Routing
-  Push-based

DISTRIBUTED LOG



 Kafka

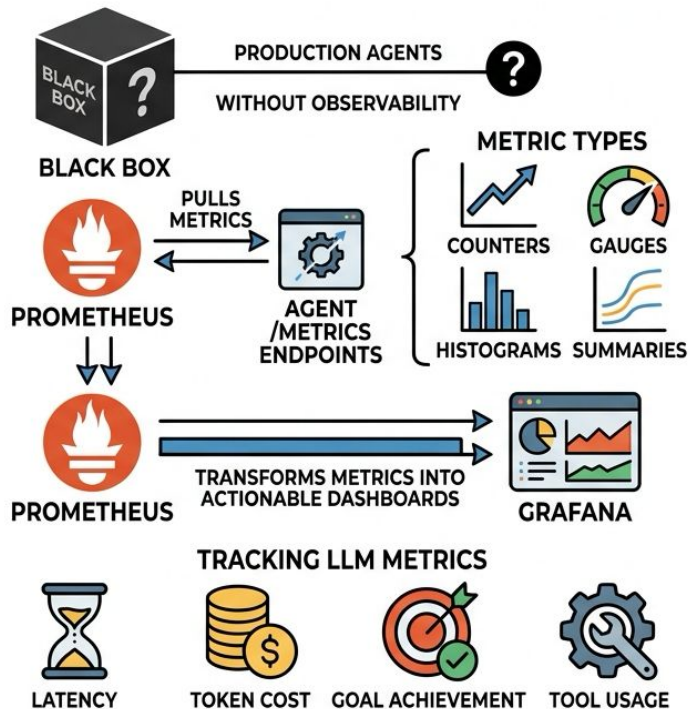
-  High-throughput Streaming
-  Replayable Event History

Vector Databases -- Semantic Memory at Scale

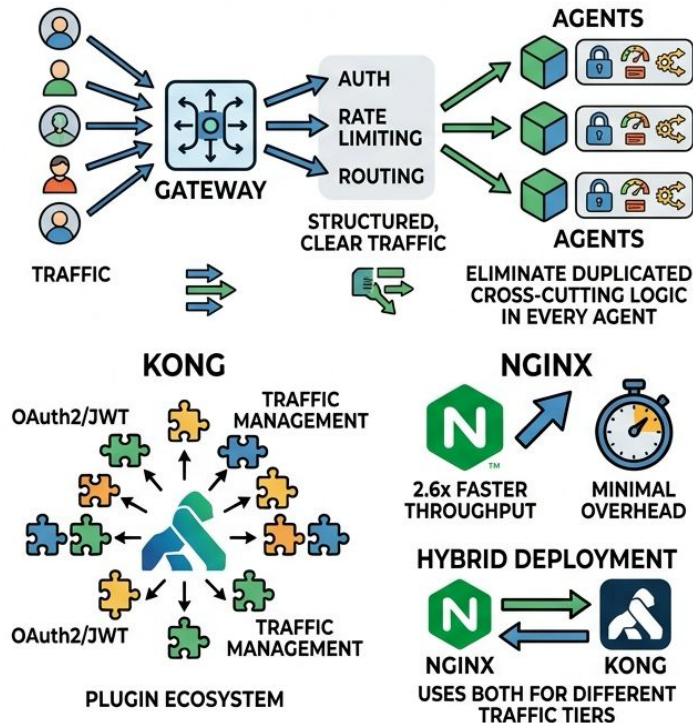
- Agents need semantic search, not just keyword matching
 - Enables RAG, long-term memory, and contextual retrieval
 - Pinecone: fully managed, fastest path to production
 - Weaviate: open-source, hybrid search, deploy anywhere
 - Milvus: maximum throughput, GPU-accelerated, multi-tenant
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Observability & Traffic Control

OBSERVABILITY WITH PROMETHEUS AND GRAFANA

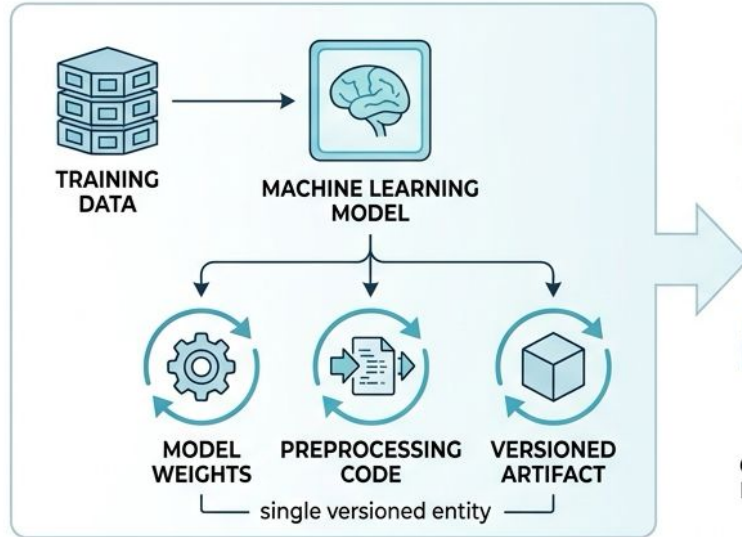


API GATEWAYS: CENTRALIZED TRAFFIC CONTROL

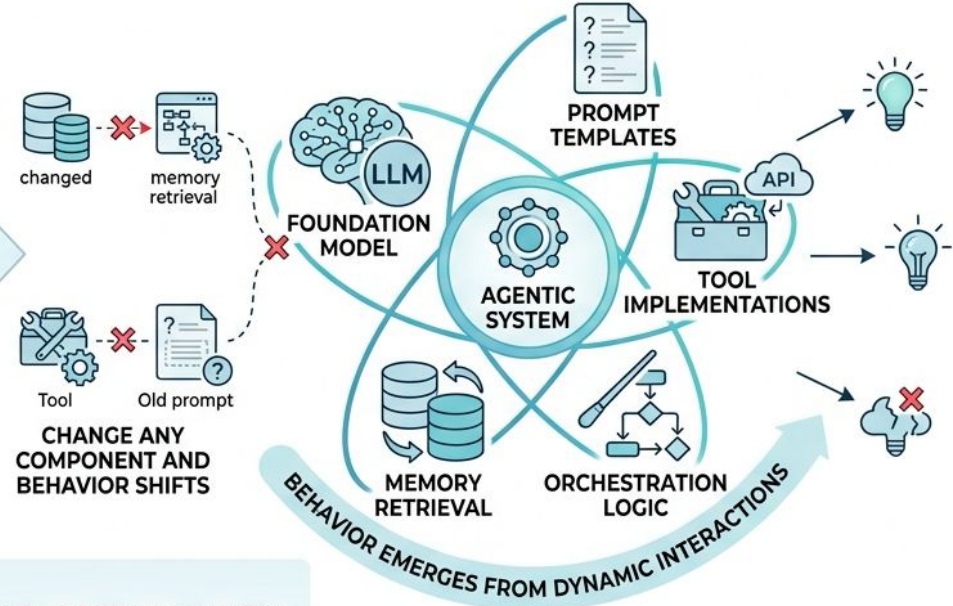


The Prototype-to-Production Gap

TRADITIONAL MLOps: THE SINGLE ARTIFACT ASSUMPTION



AGENTIC MLOps: THE COMPONENT CONSTELLATION

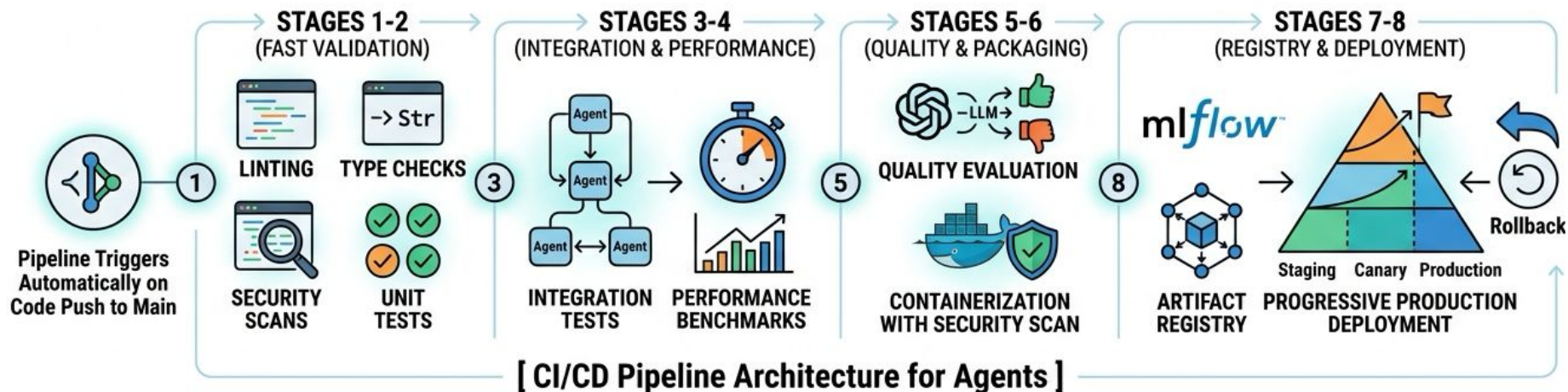


AGENTIC MLOps IS THE COORDINATION
OF THE ENTIRE ECOSYSTEM

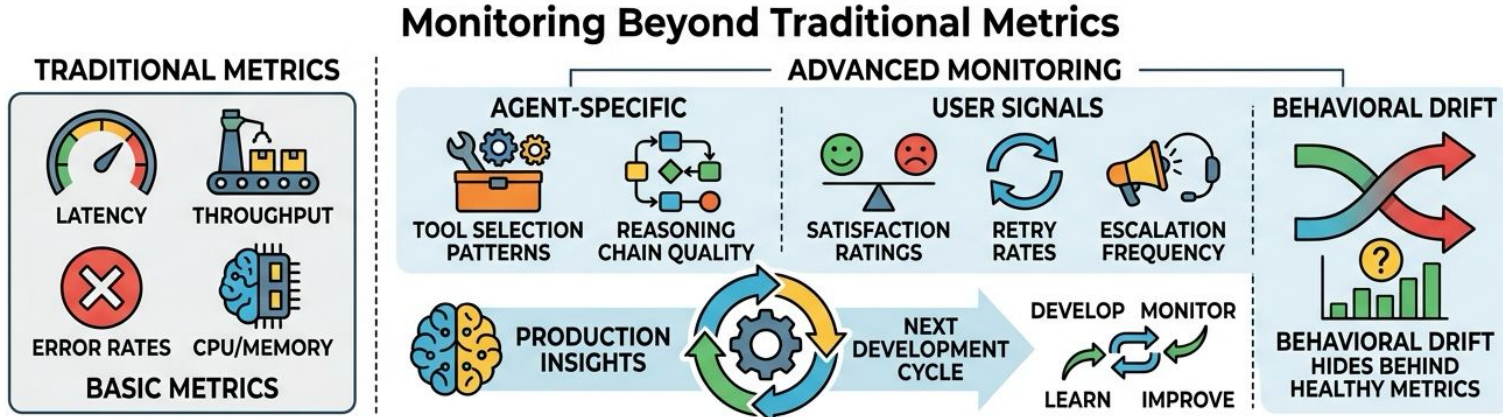
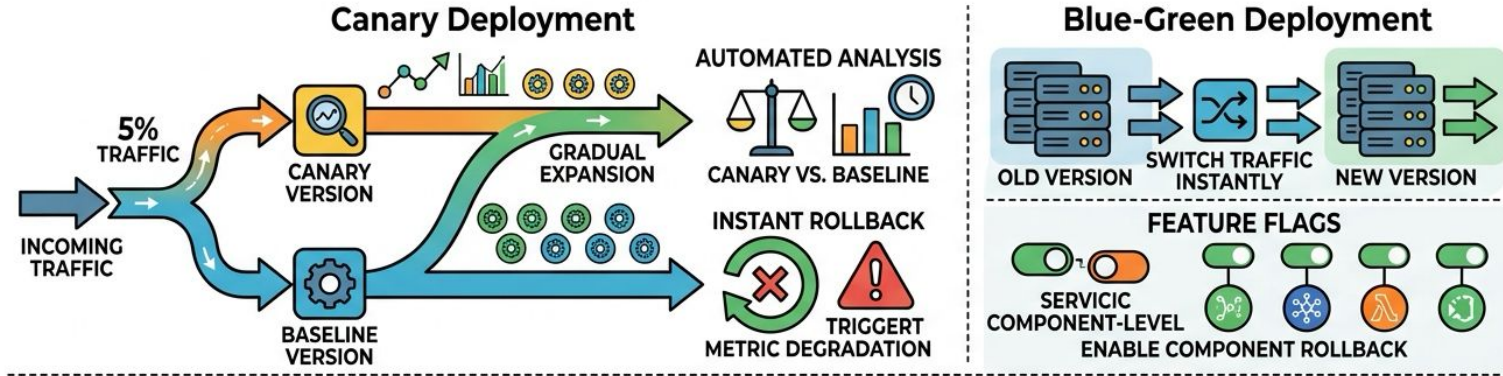
The Five Principles of Agentic MLOps

1. Automation: every step from commit to production, no manual gates
 2. Version everything: prompts, tools, memory schemas, not just code
 3. Test continuously across unit, integration, performance, quality
 4. Monitor beyond latency -- track tool patterns and reasoning drift
 5. Roll back safely with blue-green or canary strategies
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CI/CD Pipeline Architecture for Agents



Progressive Deployment & Monitoring

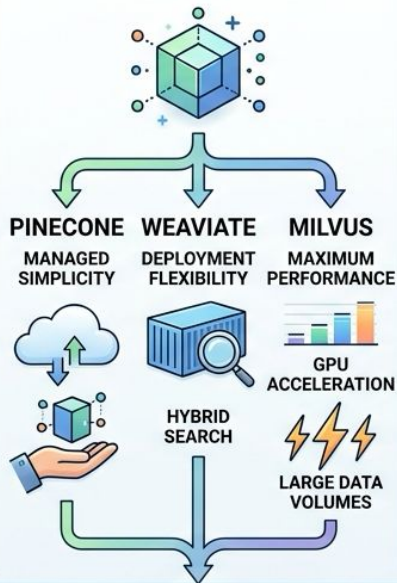


Infrastructure Decision Framework

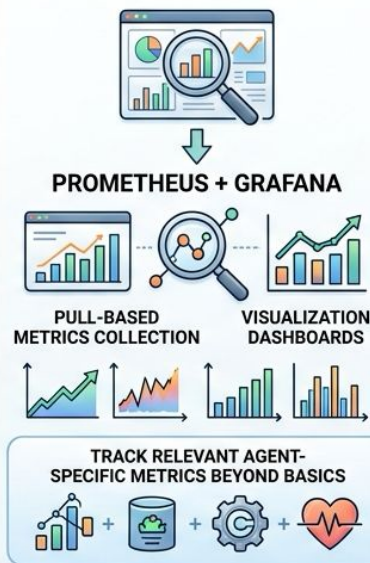
MESSAGE QUEUES



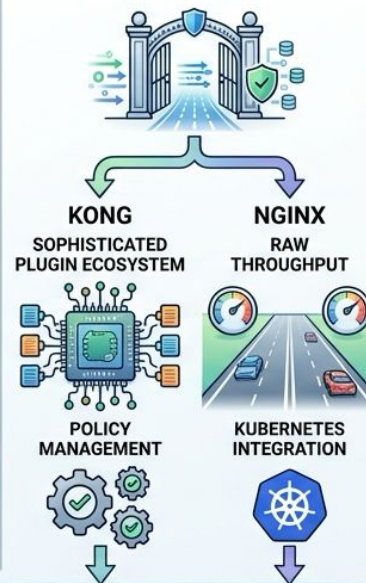
VECTOR DATABASES



OBSERVABILITY



API GATEWAYS



MATCH INFRASTRUCTURE TO TEAM EXPERTISE, NOT ASPIRATIONS



From Infrastructure to Implementation

- This chapter mapped the deployment infrastructure landscape
 - Chapter 4.2 dives into microservices and serverless patterns
 - Container orchestration with Kubernetes follows in 4.3
 - Performance profiling, NIM, TensorRT-LLM, and horizontal scaling ahead
 - The infrastructure you just learned powers everything that follows
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